

In Vitro Synthesis of Whitlockite at Physiologically Relevant Conditions

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Background

- Pathological calcification is the irreversible buildup of calcium phosphates (CaP) in soft tissues such as the medial layer of the arteries¹ and cartilage².
- These CaPs consist of phases such as dicalcium phosphate dihydrate (DCPD)^{1,2,3}, hydroxyapatite (HAP)^{1,2,3} and whitlockite (WL).^{1,2,4}

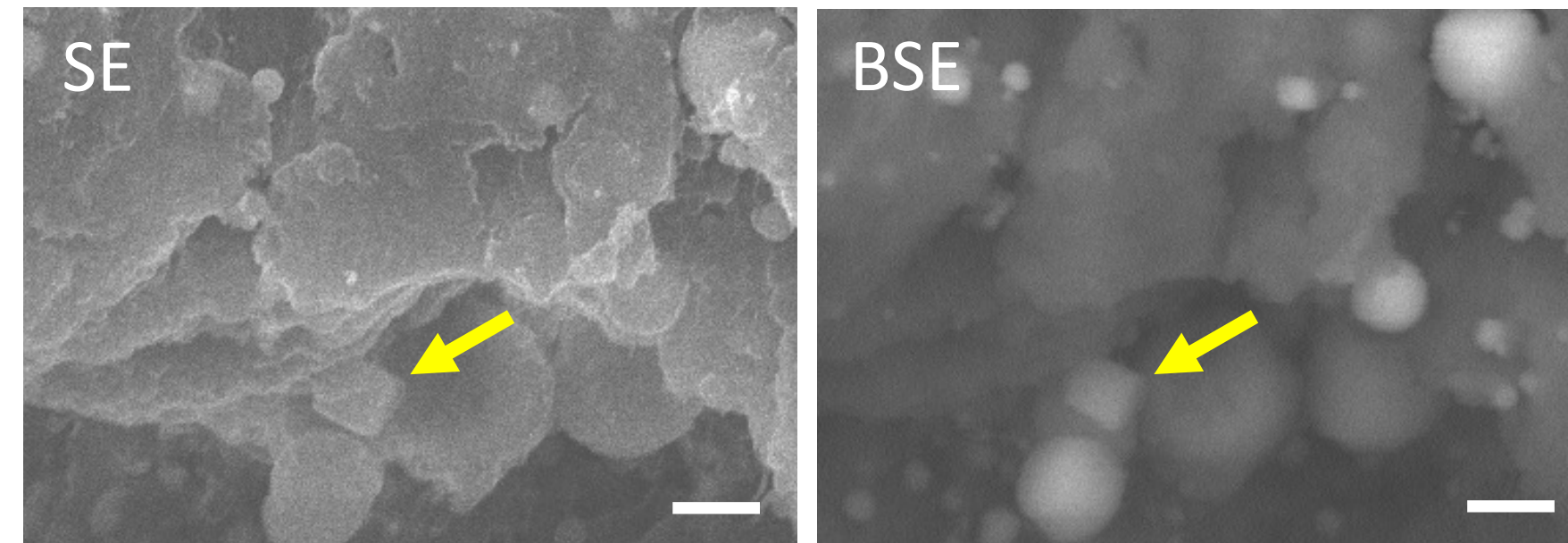


Fig. 1. SEM images (scale bars = 1 μm) of human aortic valve with WL cubic morphology. SE: secondary electron; BSE: backscattered electron.

- Despite being prevalent in the body, the magnesium-substituted CaP WL has not been produced under physiological conditions in the lab.
- The two primary WL *in vitro* synthesis methods:
 - DCPD transformation in magnesium concentrated solution at 37°C.^{5,6}
 - Direct precipitation from Ca^{2+} , Mg^{2+} , and PO_4^{3-} precursor salts in solution at high temperatures.^{7,8,9}

Objective

To synthesize WL *in vitro* at physiological conditions to better understand the formation of whitlockite in the body.

- Develop thermodynamics model to determine optimal synthesis conditions
- Design and implement experiment for synthesis of WL at physiologically relevant conditions

Methods: Mathematical Model

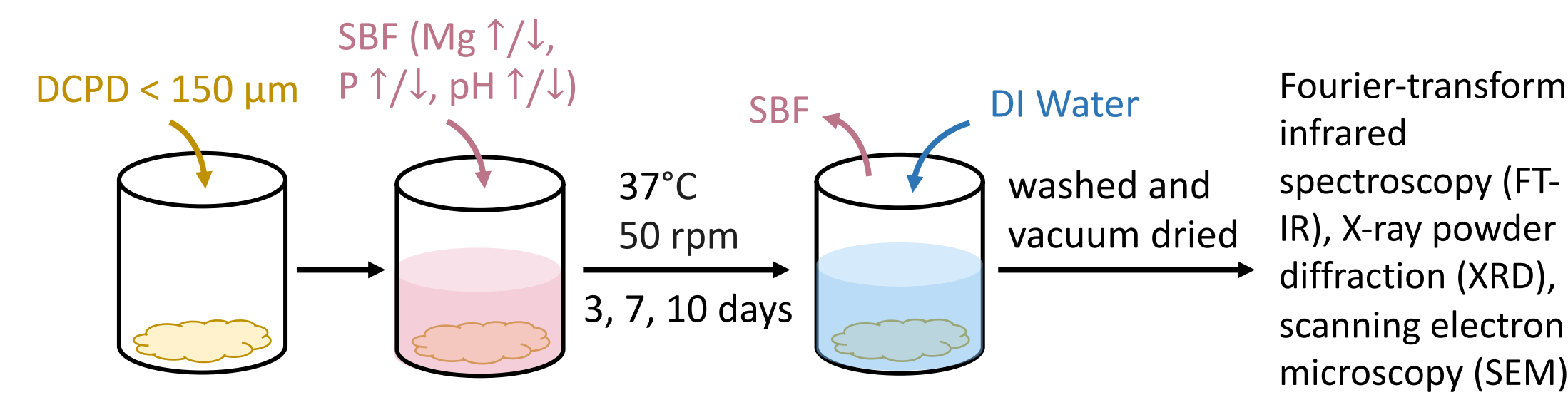
We modeled Gibbs free energy of CaP phases using:

$$\Delta G = -\frac{RT}{N} \ln \frac{IP}{K_{sp}}$$

R = ideal gas constant; T = absolute temperature;
N = number of ions; IP = ionic activities product;
 K_{sp} = solubility product.

HAP $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ $K_{sp} = 2.12 \times 10^{-59} \text{ M}^9$ [10]	DCPD $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ $K_{sp} = 2.19 \times 10^{-7} \text{ M}^2$ [10]	WL $\text{Ca}_9\text{Mg}(\text{HPO}_4)(\text{PO}_4)_6$ $K_{sp} = 7.41 \times 10^{-109} \text{ M}^{17}$ [11]*
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Methods: WL Synthesis



Simulated body fluid (SBF) is a solution with an ion concentration similar to that of human plasma¹².

Thermodynamics Model of CaP Phases

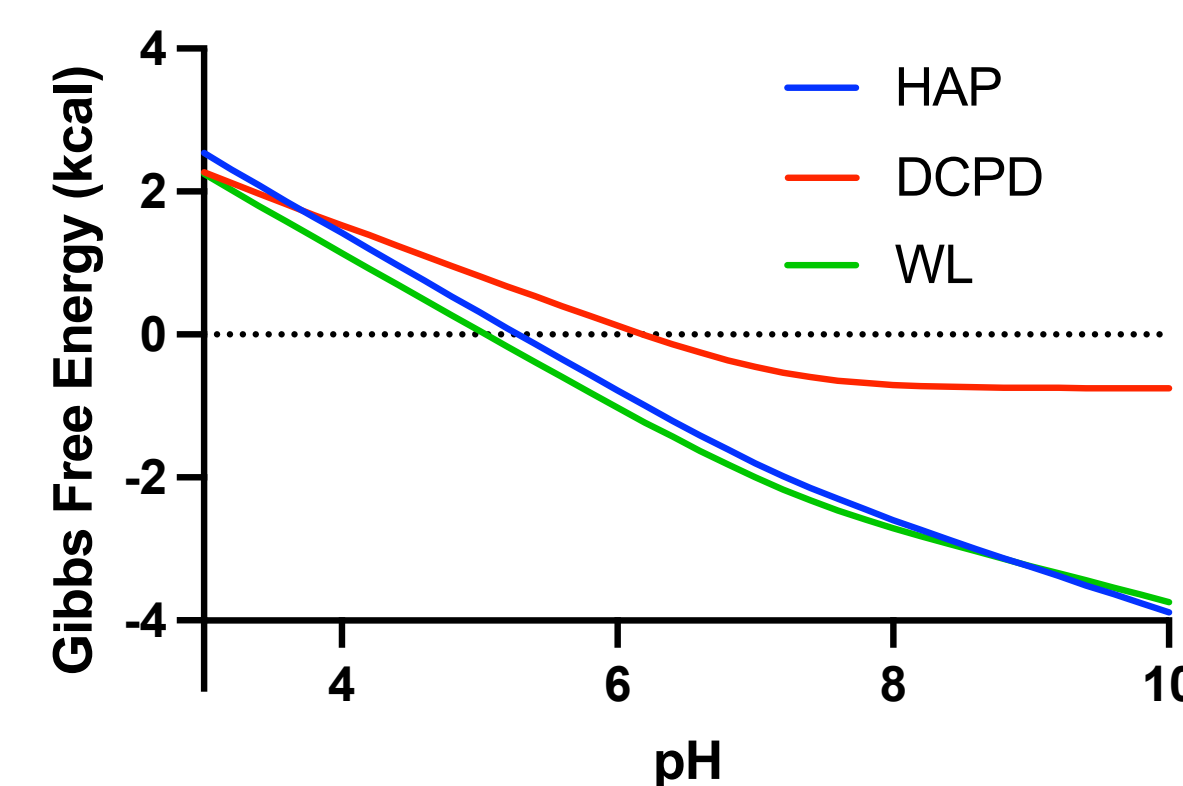


Fig. 2. As pH decreases, WL formation is favored over HAP.

$[\text{Ca}^{2+}] = 1.0 \text{ mM}$
 $[\text{Mg}^{2+}] = 1.5 \text{ mM}$
 $[\text{P}] = 1.0 \text{ mM}$

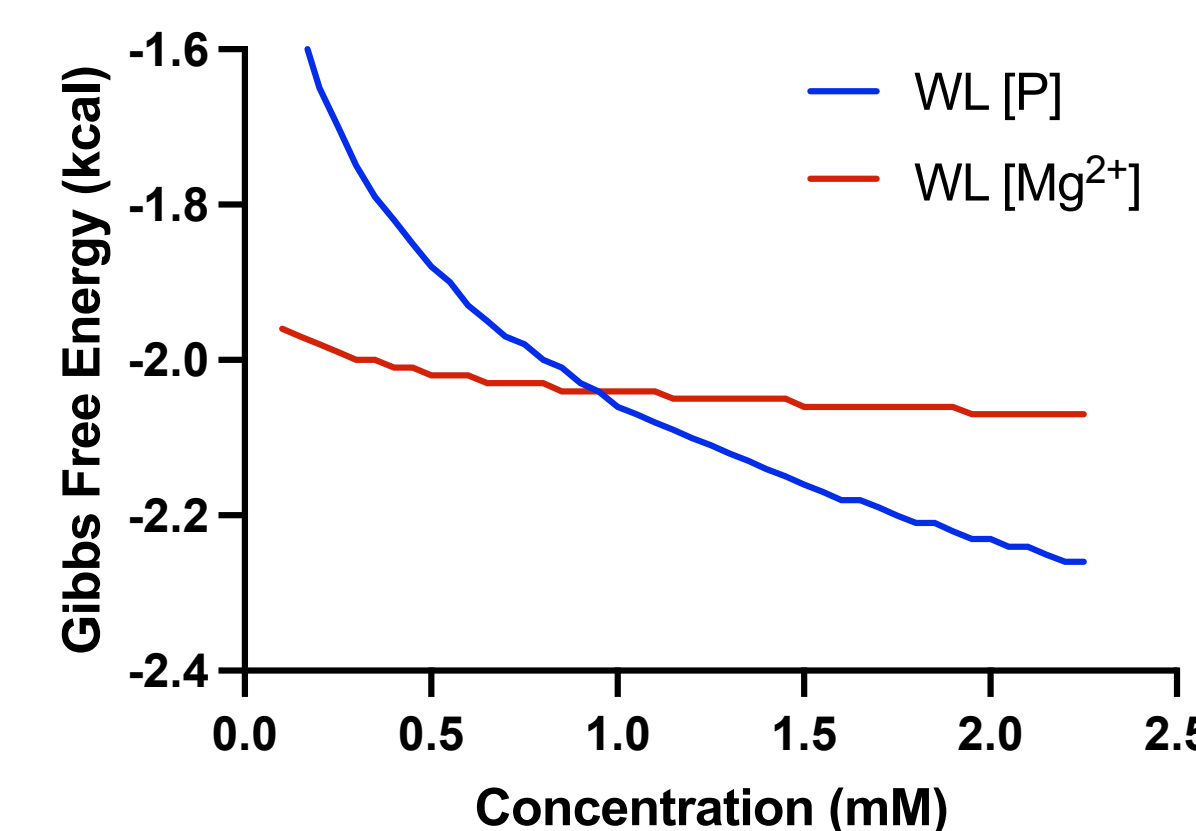


Fig. 3. Decreasing Ca/P ratio has greater impact than Ca/Mg ratio on the favorability of WL formation.

pH = 7.4
 $[\text{Ca}^{2+}] = 1.0 \text{ mM}$

WL Synthesis DoE

Run	Ca/Mg	Ca/P	pH	Time (days)
1	1.67	1.67	6.2	7
2	4	2.1	6.8	7
3	9	1.67	6.8	3
4	1.67	2.5	6.8	10
5	4	2.5	6.2	3
6	9	2.1	6.2	10
7	1.67	2.1	7.4	3
8	4	1.67	7.4	10
9	9	2.5	7.4	7

Table 1: WL synthesis runs for Screening Design DoE (n=3).

Characterization Techniques: FT-IR, XRD, and SEM

Results: FT-IR, XRD, and SEM Characterization

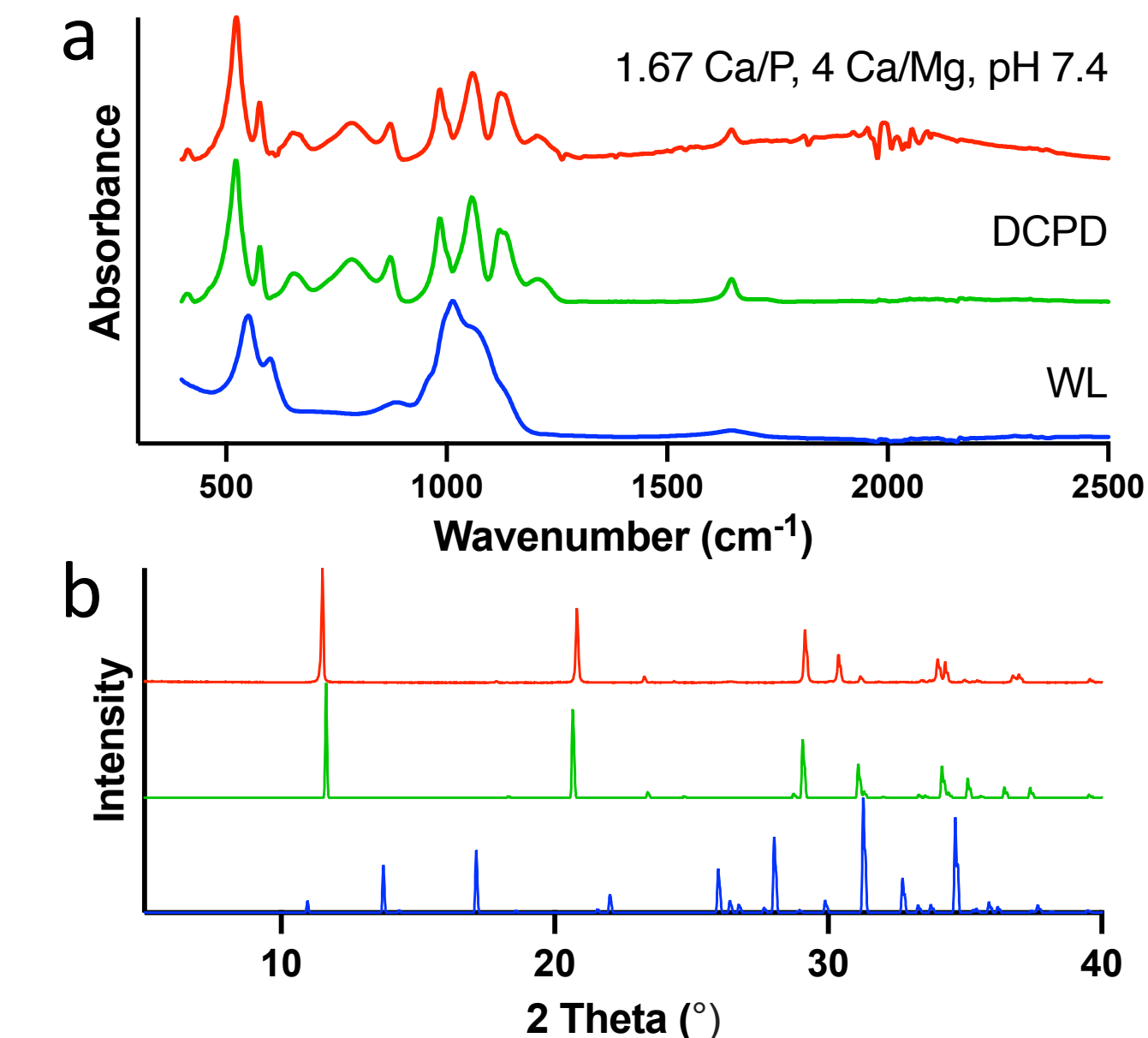


Fig. 4. (a) FT-IR and (b) XRD results with a representative Ca/P, Ca/Mg, and pH condition immersed for 10 days. When compared to exemplary DCPD and WL spectra, the results closely resemble DCPD.

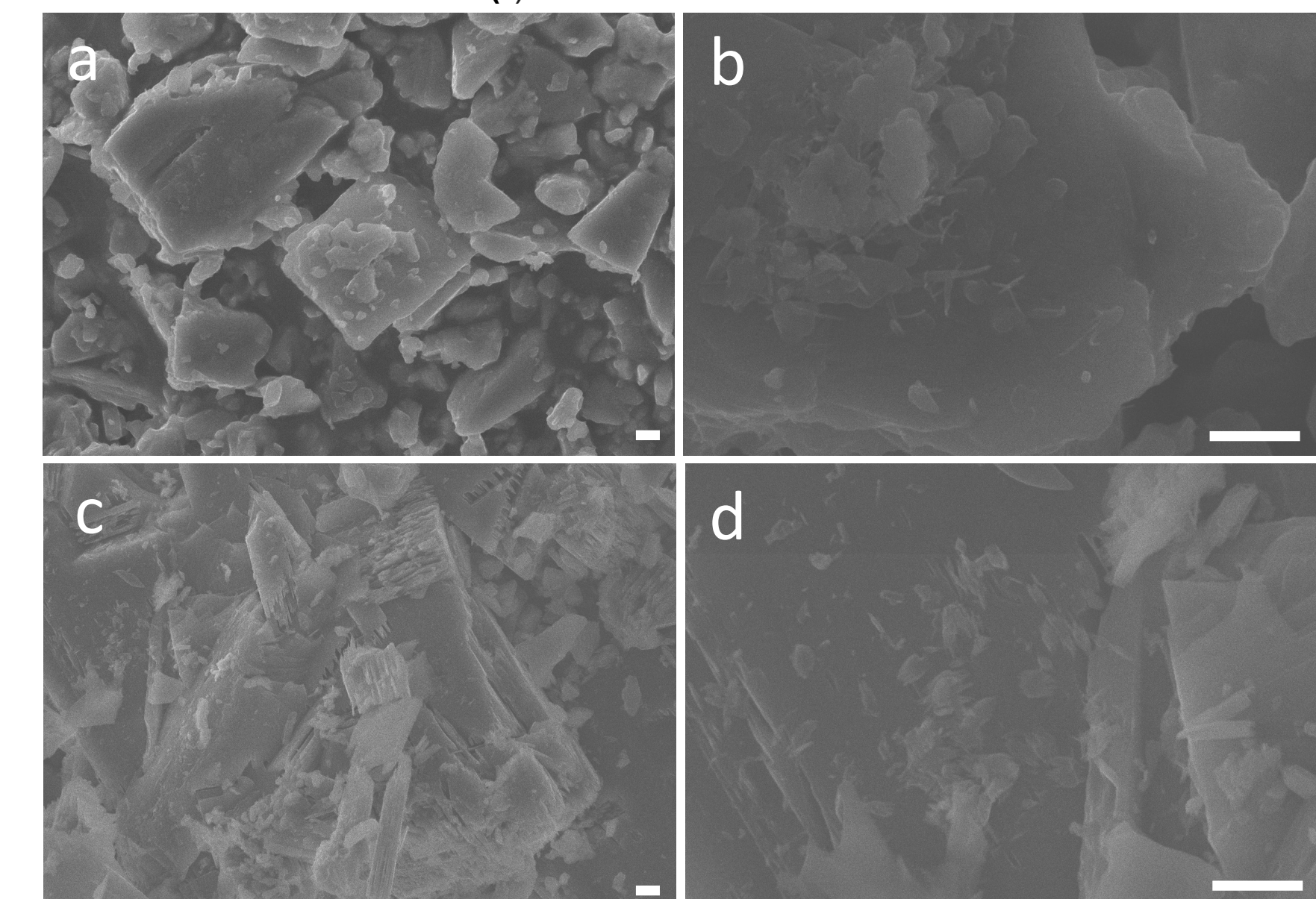


Fig. 5. Representative SEM images (scale bars = 1 μm) of DCPD (a-b) before immersion and (c-d) after immersion. We observed characteristic signs of dissolution.

Conclusions/Future Work

Further research is required to synthesize WL at physiological conditions. Next steps include using Mg-substituted DCPD, increase time immersed > 20 days, and altering the precursor's crystallinity. To develop more physiologically relevant models, synthesizing WL from DCPD in collagen and/or elastin gels should be explored.

Acknowledgements

